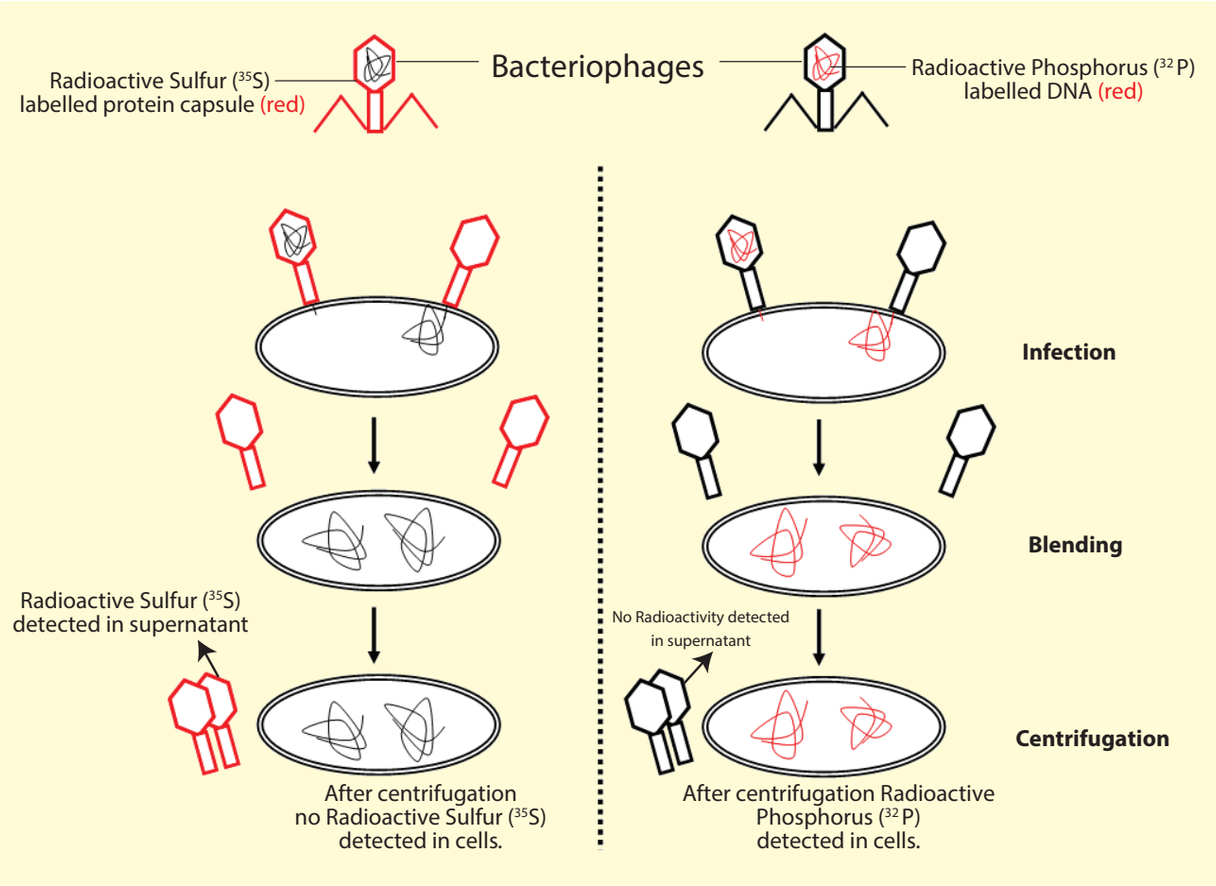


Fig. 5.2 The Hershey-Chase (blender) experiment

5.3 DNA is the genetic material

Many biologists despite the earlier experiments of Griffith, Avery and others, still believed that protein, not DNA, was the hereditary material in a cell. As eukaryotic chromosomes consist of roughly equal amounts of protein and DNA, it was said that only a protein had sufficient chemical diversity and complexity to encode the information required for genetic material. In 1952, however, the results of the Hershey-Chase experiment finally provided convincing evidence that DNA is the genetic material.

5.3.1 Hershey and Chase experiment on T_2 bacteriophage

Alfred Hershey and **Martha Chase** (1952) conducted experiments on bacteriophages that infect bacteria. Phage T_2 is a virus that infects the bacterium *Escherichia coli*. When Phages (virus) are added to bacteria, they adsorb to the outer surface, some material enters the bacterium, and then later each bacterium lyses to release a large number of progeny phage. Hershey and Chase wanted to observe whether it was DNA or protein that entered the bacteria. All nucleic acids contain phosphorus, and proteins contain sulphur (in the amino acid cysteine and methionine). Hershey and Chase designed an experiment using radioactive isotopes of Sulphur (^{35}S) and phosphorus (^{32}P) to keep separate track of the viral protein and nucleic acids during the infection process. The phages were allowed to infect bacteria in culture medium which containing the radioactive isotopes ^{35}S or ^{32}P. The bacteriophage that grew in the presence of ^{35}S had labelled proteins and bacteriophages grown in the presence of ^{32}P had labelled DNA.

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